CareerArth AI

CAREERARTH AI · ENTREPRENEURSHIP CAPSTONE

Final Capstone Report

A structured, AI-assisted career diagnostic as the entry point to human career advisory — built, tested, and honestly bounded.

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Course Entrepreneurship Capstone (ENT-CP-4997-1)

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Evidence key (used throughout this report)

Because this capstone contains both a working product and an untested business plan, every material claim in this report belongs to one of three categories, marked where it matters:

- **[DEMONSTRATED]** — verifiable in the repository today: running code, passing tests, logged output.
- **[IMPLEMENTED, UNVALIDATED]** — the code exists and is complete, but has never been exercised under real conditions.
- **[HYPOTHESIS]** — a reasoned business assumption with no supporting evidence yet, awaiting the proposed pilot.

No real customer has used this product. No revenue, conversion, or willingness-to-pay data exists. Nothing in this report should be read as claiming otherwise.

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1. Executive Summary

CareerArth is an existing career-advisory brand whose website funnels visitors from a free "5-minute assessment" promise toward paid human consultation. This capstone asks a focused entrepreneurial question: **can an AI-assisted diagnostic make that funnel genuinely valuable at the top — good enough that a stranger would complete it, trust it, and want to talk to a consultant afterwards — without the business overclaiming what the AI knows?**

The project delivers four things:

1. **A working MVP** [DEMONSTRATED]: a branded web product where a user completes a 5–7 minute structured intake, receives a deterministic ARTH scorecard (Alignment, Risk Exposure, Trajectory, Human Capital), and gets a personalised career analysis with matched career paths and a 90-day plan — followed by a consultant review workflow and full funnel instrumentation. 42 of 42 automated tests pass.
2. **A business model recommendation** [HYPOTHESIS]: use the free diagnostic as lead-generation for paid consultation (₹1,999–₹4,999 hypothesised price range), because modelling shows consultant time — not AI cost (estimated $0.05–$0.15 per report) — is the dominant variable cost, and because this formalises a motion the production CareerArth site already runs.
3. **An honest evidence audit**: six synthetic personas demonstrate the system differentiates inputs (overall scores from 42 to 88 across four interpretation bands), but all generated reports to date use a deterministic mock generator — no real Claude-generated report has ever been produced or read, and no real user has touched the product.
4. **A ready-to-run validation kit**: a 10–20-person pilot protocol with recruitment message, consent text, surveys, interview guide, willingness-to-pay questions, and explicit success/failure thresholds — executable after submission with zero additional engineering, because the MVP's instrumentation already measures every funnel step the pilot needs.

The honest bottom line: **technical feasibility is demonstrated; commercial desirability is deliberately left as a designed, falsifiable experiment.** The most important number in this venture

— the rate at which a free diagnostic converts a stranger into a paying consultation client — cannot be known from a laptop, and this report does not pretend to know it.

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2. CareerArth Context and the Problem

2.1 The company context

CareerArth operates a live website offering career diagnostics and strategic advisory. Its public framing is distinctive: careers fail not through dramatic events but through quiet erosion — "gilded stagnation," where a well-paid professional's options narrow invisibly while their salary suggests everything is fine. The site offers a free assessment and routes interested visitors to a consultation page ("Talk to a Career Expert") with no published pricing. The ARTH framework — **A**lignment, **R**isk Exposure, **T**rajectory, **H**uman Capital — is the house methodology, with a published sample scorecard.

This capstone was built as a strictly isolated prototype (`careerarth-ai/` in the same repository). The production website was not modified.

2.2 The customer problem

Mid-career professionals who have stopped growing face a specific, uncomfortable epistemic problem: **they can feel that something is wrong but cannot decompose it.** Is the problem their skills? Their industry? Their network? Their own ambition? The feeling is diffuse; the available responses are extreme (do nothing, or quit) or expensive (hire a coach before knowing whether the problem warrants one).

Their current options each fail in a characteristic way [HYPOTHESIS — reasoned from the alternatives analysis, not user-researched]:

- **Asking ChatGPT** produces fluent advice with no structure, no grounding in their actual profile, and no accountability.
- **Career coaches** are effective but demand a large, vague commitment upfront — precisely when the customer doesn't yet know if their problem justifies it.
- **LinkedIn** shows them a feed, not a decision.
- **Personality assessments** describe traits, not their situation.
- **Free content** requires them to do all the synthesis themselves.

2.3 The entrepreneurial opportunity

The opportunity is not "AI career advice" — that is free and abundant. The opportunity is narrower and more defensible as a *business motion*: **an inexpensive, structured, trustworthy first step that sits between "vague worry" and "expensive commitment,"** owned by a brand that already sells the expensive commitment. The diagnostic's job is not to be the product; its job is to convert diffuse anxiety into a specific, named problem — at near-zero marginal cost — and hand a well-briefed, high-intent client to a human consultant whose time is the scarce, monetisable asset.

This reframing matters because it survives the strongest objection ("ChatGPT does this for free"): a generic chatbot can produce advice, but it cannot produce a structured, consistent intake, a deterministic score, a consultant-ready brief, and a booked consultation for a firm the customer already half-trusts.

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3. Initial Customer: ICP and Job-To-Be-Done

3.1 Beachhead segment [HYPOTHESIS]

Mid-career professionals (5–10 years of experience) experiencing career stagnation — no promotion or material scope change in 2+ years, currently employed, in the technology, finance, or marketing/consulting footprint the product's knowledge base actually covers.

Three candidate segments were evaluated in Stage 3. Early-career professionals (1–3 years) have real but diffuse pain and the weakest willingness-to-pay case. Role/industry switchers are high-intent but stress the product's shallowest asset — an 18-role knowledge base — where a switcher's need is breadth. The stagnation segment was chosen because it matches what the product measures *best*: the ARTH rubric contains an explicit, deterministic tenure penalty (5+ years in role reduces the Trajectory score), and the persona built around this pattern produces the most legible diagnostic tension in testing — strong Human Capital (76) against Critical Trajectory (23), which reads as "you have assets; you are not using them to move." Stagnation is also the exact pain the production CareerArth brand already markets to.

3.2 Job-to-be-done

> *"When I notice I've stopped growing but can't tell whether the problem is me, my role, or my industry, help me get an honest, structured read on where I actually stand — and a credible next move — without requiring a big commitment before I know it's warranted."*

Triggers: a denied or bypassed promotion; a second consecutive "solid" (not "exceptional") review; a peer's title change on LinkedIn; a tenure milestone.

Willingness-to-pay hypothesis [HYPOTHESIS]: this segment will not pay meaningfully for the diagnostic itself (they cannot yet judge its quality) but will pay for a human consultation *after* the diagnostic has demonstrated specificity about their situation. This is the single most important untested assumption in the venture.

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4. The Solution: CareerArth AI

4.1 Product flow

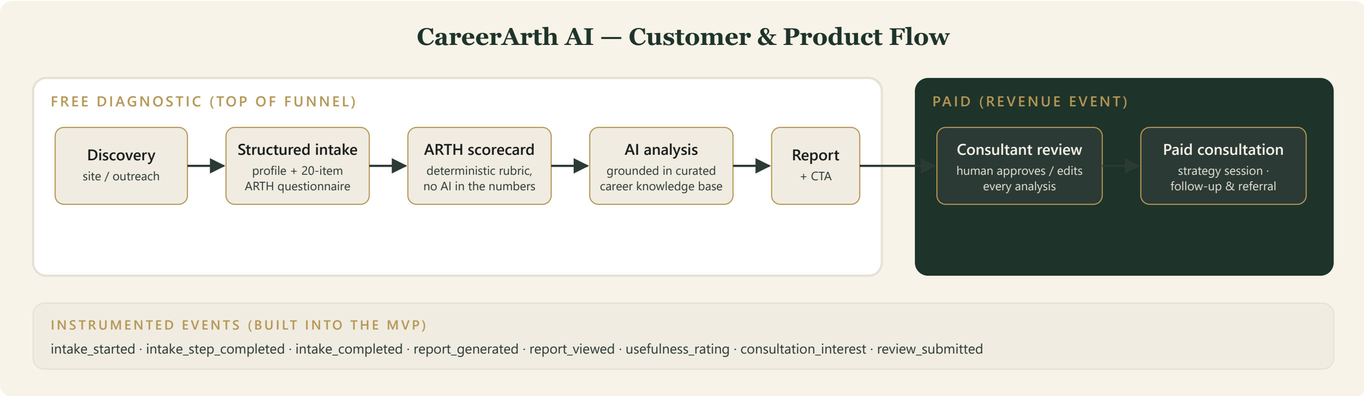

Figure 1 — Customer & product flow: free diagnostic funnel feeding the paid consultation, with the MVP's instrumented events.

The customer experience, as actually built [DEMONSTRATED]:

1. **Structured intake (5–7 min):** a branded wizard collects a compact profile (role, industry, experience, skills, goals, constraints, primary concern) and a 20-item ARTH questionnaire — five items per dimension, each answered 0–4 against anchored endpoints.
2. **Deterministic ARTH scorecard:** fixed weights and thresholds compute four dimension scores and an overall Arth Score. **No AI touches the numbers.**
3. **Personalised analysis:** the system retrieves the most relevant roles from a curated career knowledge base and drafts a structured report — overall diagnosis, dimension explanations, strengths, risks, honest tensions, 2–3 matched career paths with trade-offs, skill gaps, a 90-day action plan, and open questions for a consultant.
4. **Consultation CTA + instrumentation:** the report ends with a consultation call-to-action; every funnel step logs an event.
5. **Consultant review:** a separate review screen lets a human approve, flag, or edit each section, tracks review minutes automatically, and flips the customer-visible badge to "Consultant reviewed."

4.2 The ARTH framework — what it is and is not

ARTH organises a career situation into four questions:

Dimension	Question it answers	Weight
Alignment	Does today's work build toward the stated ambition?	0.30
Risk Exposure	How insulated is this position from industry and automation shifts? (higher = safer)	0.25
Trajectory	Are options expanding or narrowing over time?	0.20
Human Capital	How strong are the network, brand, and transferable assets?	0.25

Scores fall into five fixed interpretation bands (Strong 85–100, Stable 70–84, Watch 55–69, Vulnerable 40–54, Critical 0–39). The weighted formula reproduces the sample scorecard published on the production site (72/48/52/81 → 64) — kept in the test suite as a regression check, explicitly *not* as validation.

ARTH is a prototype structured diagnostic framework, not a scientifically validated instrument. Its weights are heuristic choices by the founding team. The system embeds this disclaimer in every generated score object and report footer, and this report's positioning (§9) treats that honesty as a feature: the product's credibility rests on structure, evidence-citation, and human review — not on claimed psychometric authority.

4.3 Why this design is an entrepreneurship decision, not just an engineering one

Three product rules were chosen specifically to make the *business* defensible:

1. **The AI never scores.** Scores come from a fixed rubric; the AI only explains them. A customer (or regulator, or professor) can ask "why is my Trajectory 23?" and receive a deterministic answer. This converts the product's weakest point (heuristic weights) from a hidden liability into an auditable, improvable asset.
2. **Every claim must cite its evidence.** The report generator is contractually required to tag each claim to a profile field, a score, or a knowledge-base entry; an automated lint rejects drafts that don't comply. This is the structural answer to "how is this better than ChatGPT?"
3. **A human reviews before anything is final.** The review layer is both a trust mechanism and the bridge to the revenue event — the consultant meets the client already briefed.

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5. The Working MVP [DEMONSTRATED]

5.1 What exists and runs

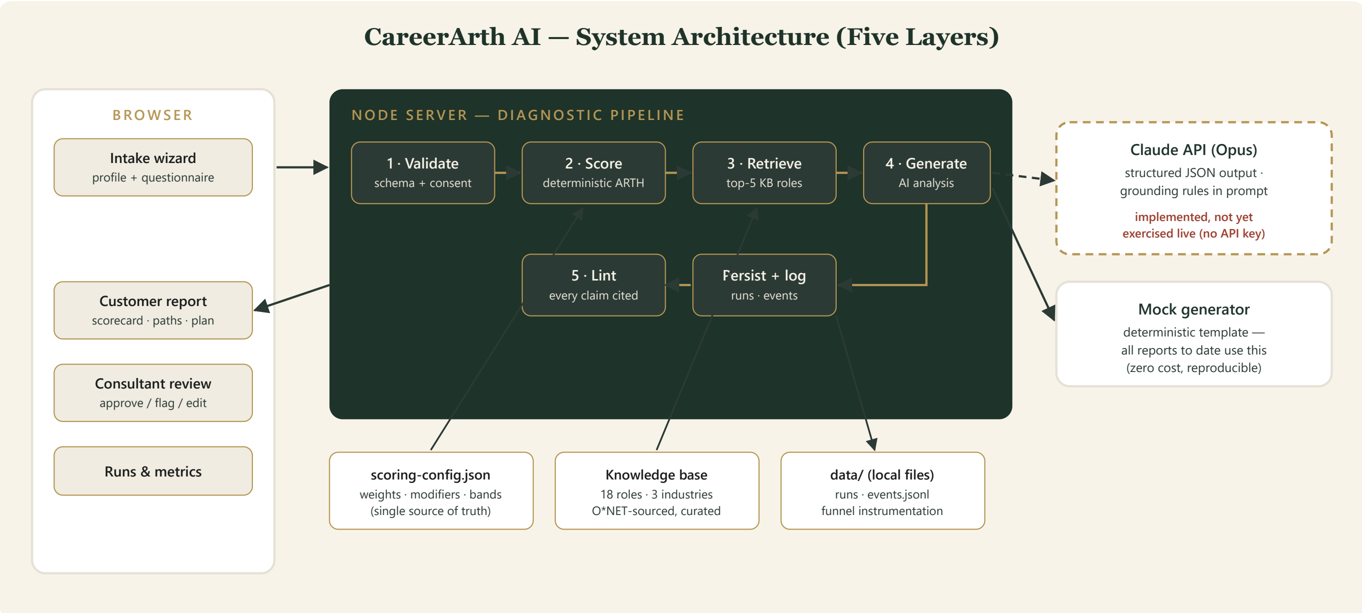

Figure 2 — System architecture: the five-layer pipeline; the Claude path is implemented but has never run live.

- Branded web intake wizard, customer report, consultant review screen, and runs index, served by a small Node server (`npm start`, localhost:4747).

- Deterministic scoring engine reading all weights, modifiers, and bands from a single configuration file, so documentation and code cannot drift apart.
- Curated knowledge base: 18 occupation profiles across technology, finance, and marketing/consulting, hand-curated from O*NET (the U.S. Department of Labor occupational database), each with core skills, transition notes, outlook, and source citation. Deterministic keyword retrieval selects the top 5 for each profile, always including the user's current and target roles when present, with a graceful out-of-corpus fallback that tells the user honestly when their field isn't covered.
- Report generation with a strict JSON structure and the citation lint described above.
- Consultant review workflow with per-section decisions, inline edits, confidence rating, and automatic time tracking.
- Instrumentation: eight event types covering the funnel from intake start to consultation interest, aggregated by a metrics endpoint.
- **42 of 42 automated tests pass** (re-verified for this report): 16 scoring tests (including the hand-calculated worked example and the sample-score calibration), 8 validation tests, 10 retrieval tests, 8 end-to-end pipeline tests.

5.1a Product walkthrough (captured from the running MVP, synthetic demo data)

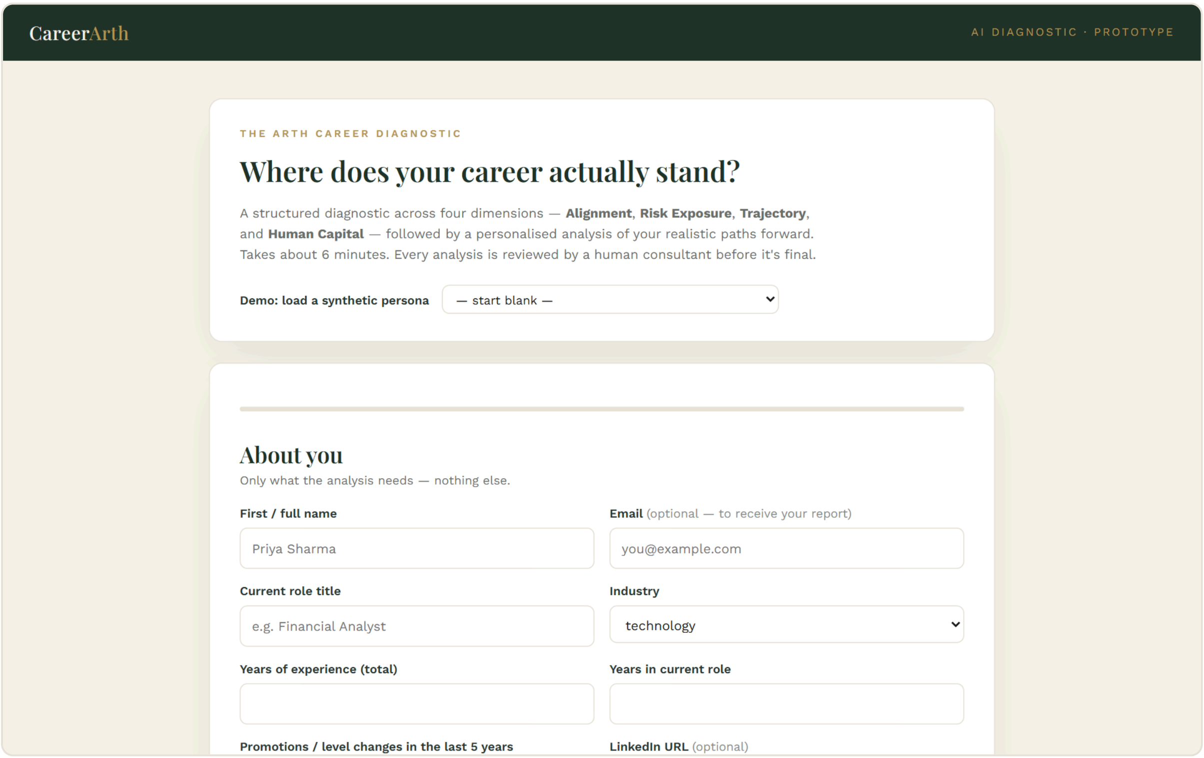

Figure 4 — Intake landing — ARTH framing, 6-minute promise, and the heuristic-score disclaimer on the page itself.

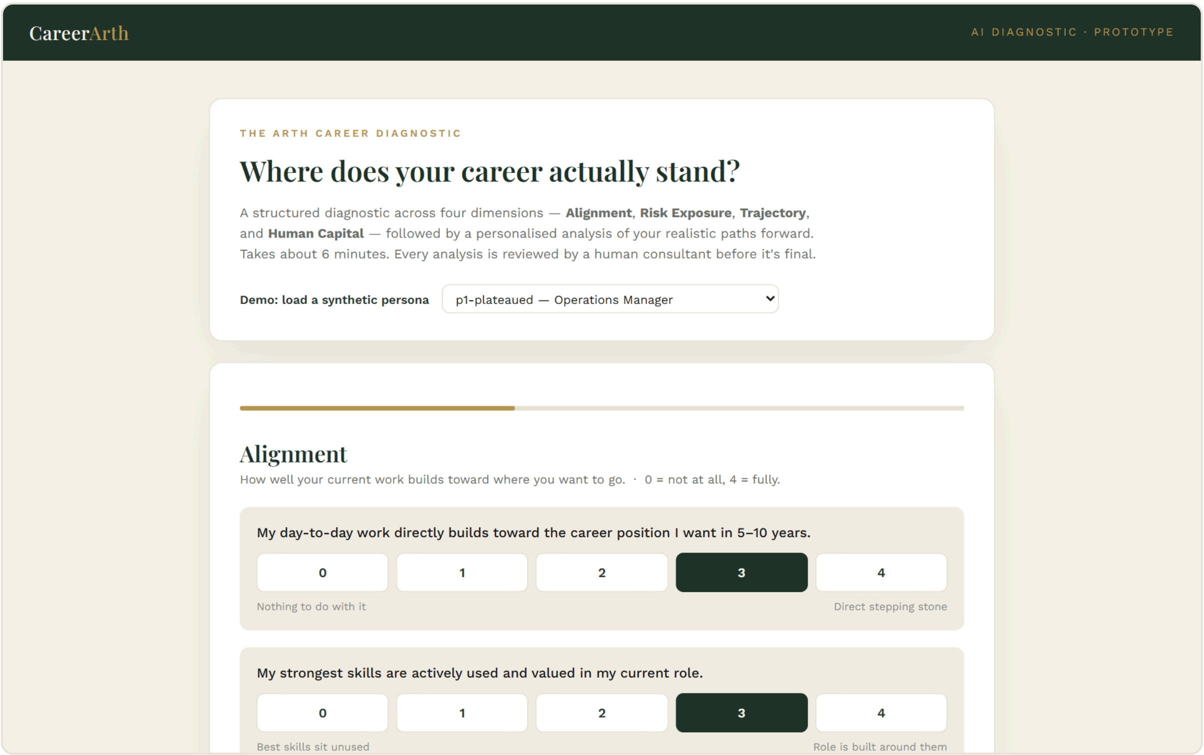

Figure 5 — Diagnostic questionnaire — anchored 0–4 items, five per dimension.

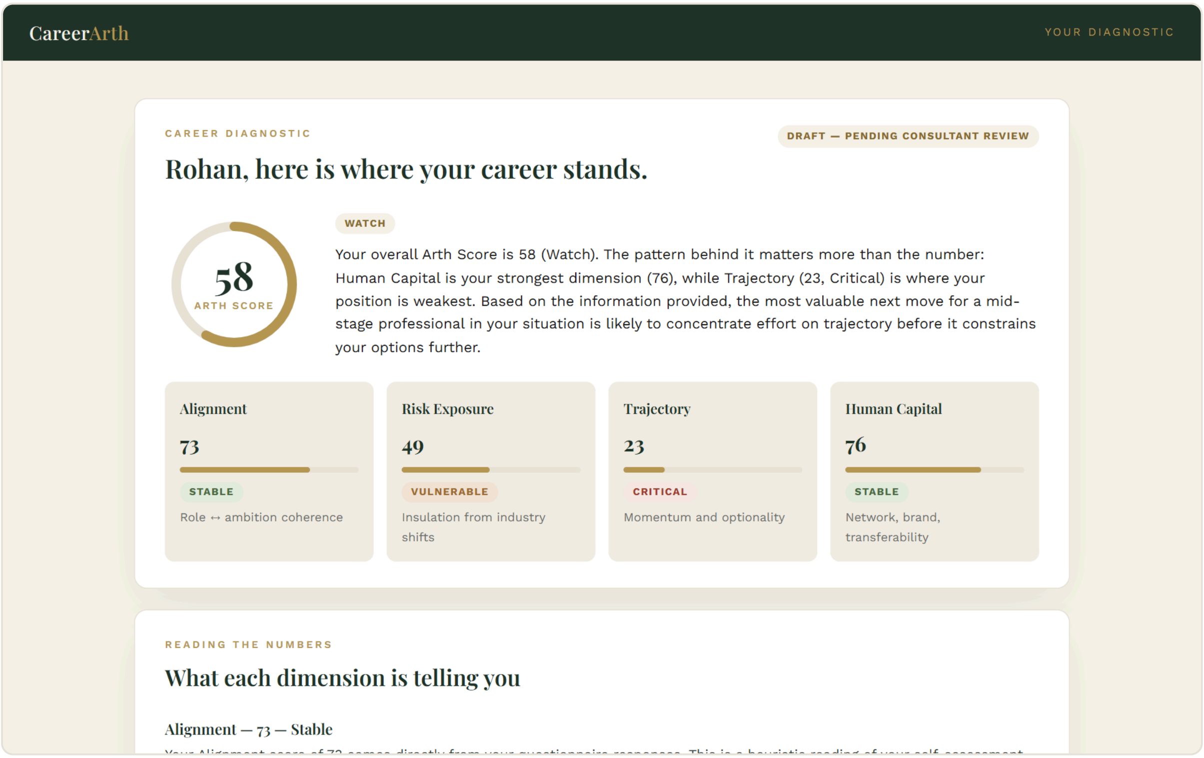

Figure 6 — The customer report: Arth Score 58 (Watch) for persona p1, with the draft-pending-review badge.

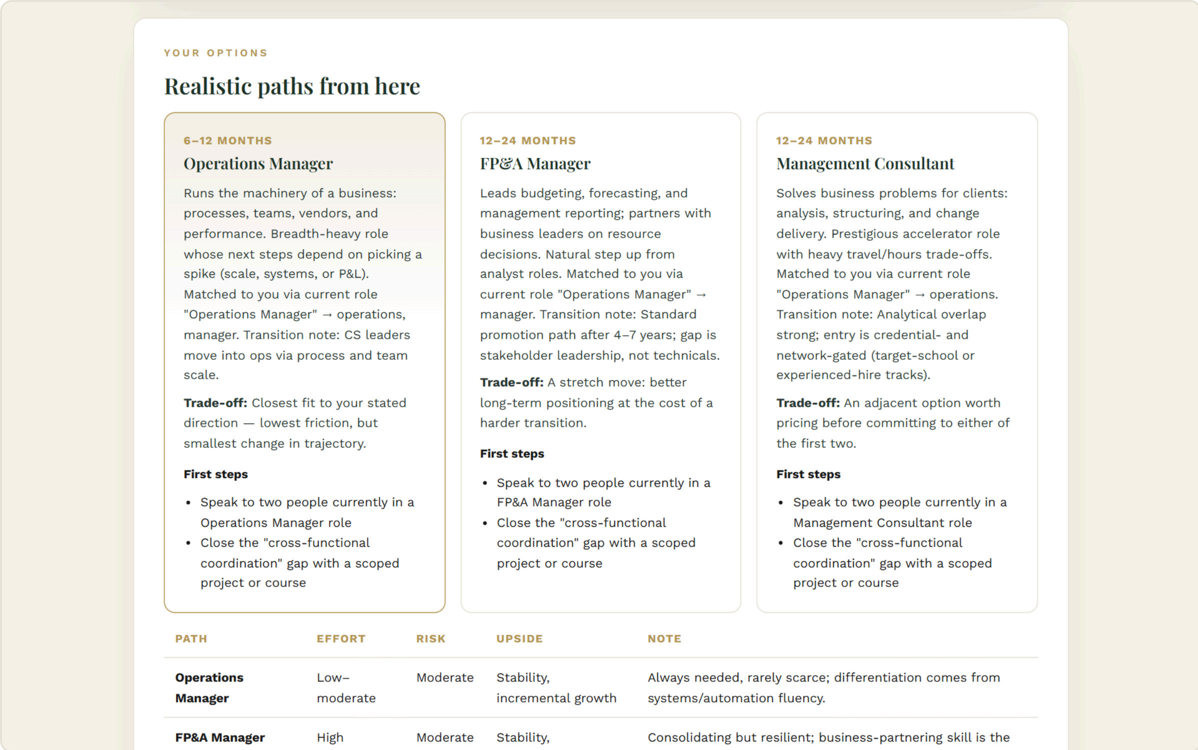

Figure 7 — Matched career paths with fit rationale and honest trade-offs.

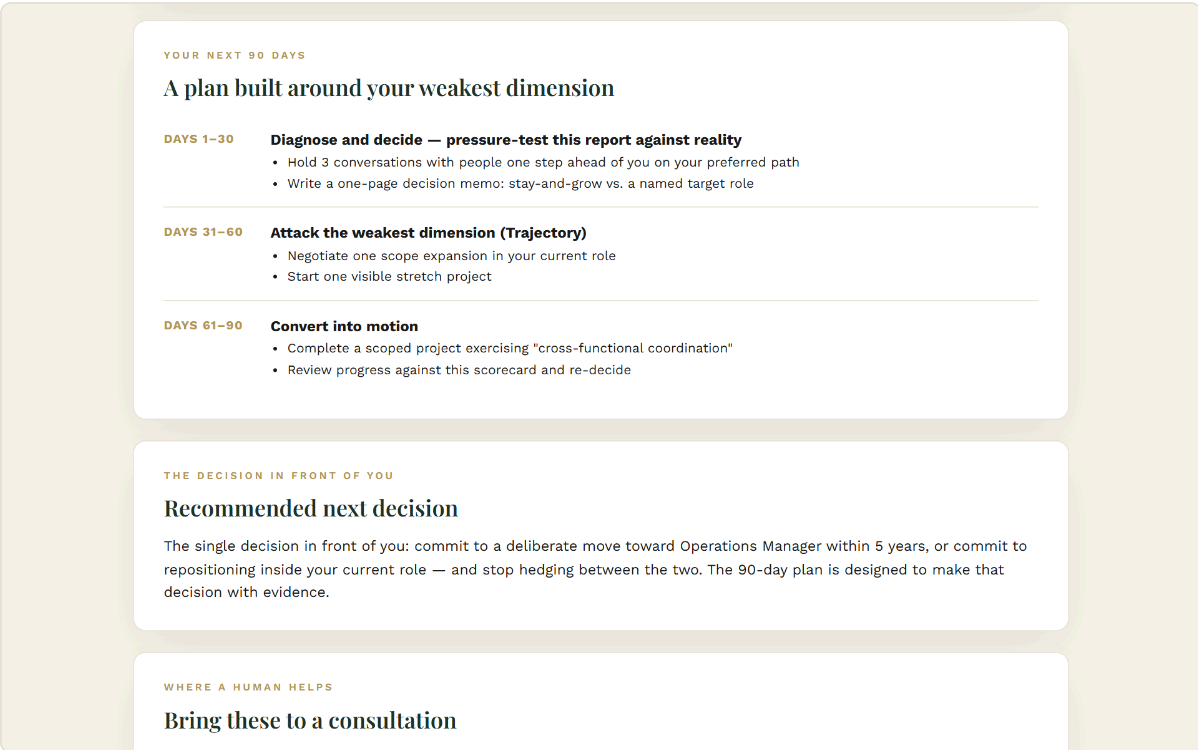

Figure 8 — The 90-day plan, built around the weakest dimension.

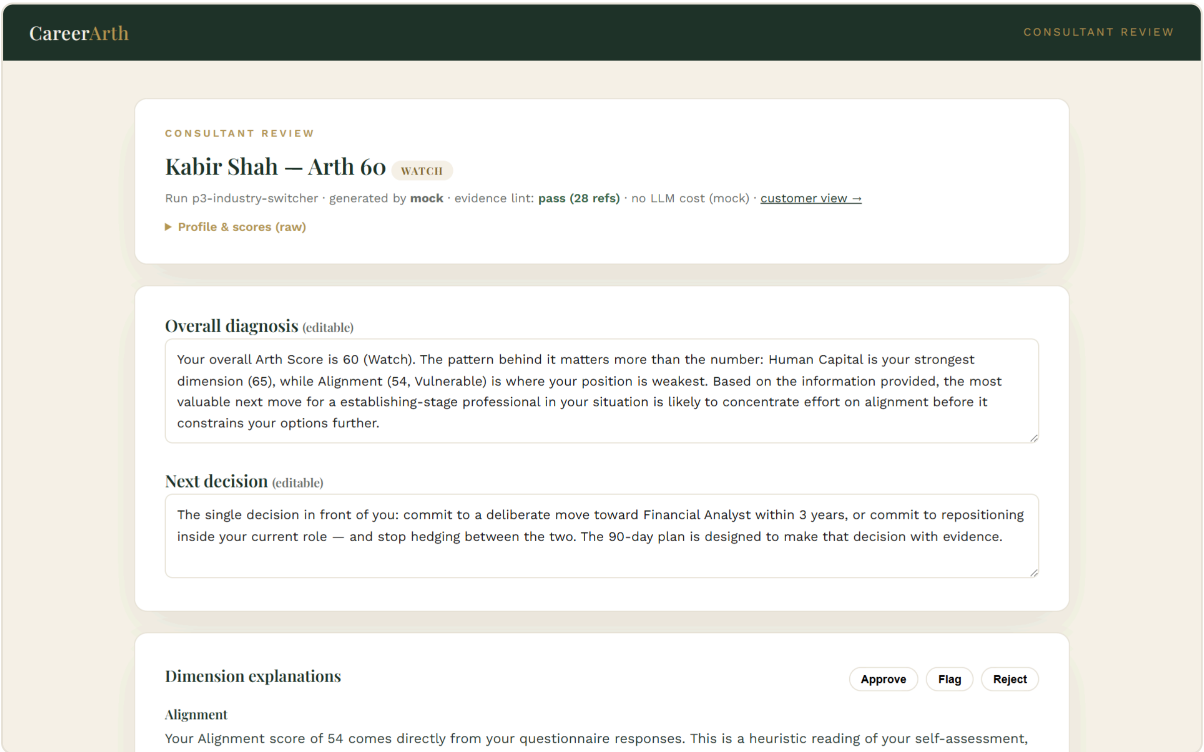

Figure 9 — Consultant review screen — per-section decisions, tracked minutes.

5.2 The Claude integration [IMPLEMENTED, UNVALIDATED]

The Claude API integration is fully written — model call, structured-output JSON schema, refusal and truncation handling, token-usage cost accounting. It has **never been executed**, because no API key was configured during the project. Every report ever generated by this system — including all six persona reports — was produced by a deterministic mock generator that fills the same report structure from the scores and retrieval results.

This is reported plainly because it bounds what the project can claim: the *pipeline* is demonstrated; the *quality of a real AI-generated report* — its tone, specificity, and groundedness under real generation — is unknown, and is the first item on the post-capstone validation roadmap (§13).

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6. Synthetic Testing and Results [DEMONSTRATED — as QA only]

Six synthetic personas were designed to stress different parts of the system. Their results, re-verified from the stored runs:

Persona	Situation (invented)	A	R	T	H	Overall	Band
p4 gilded-stagnation	Highly paid senior engineer, obsolete stack	60	34	10	55	**42**	Vulnerable
p6 returner	Re-entering after an 18-month career break	73	38	38	48	**51**	Vulnerable
p1 plateaued	Ops manager, 6 years in seat, strong network	73	49	23	76	**58**	Watch
p3 industry-switcher	Finance analyst targeting tech PM	54	64	58	65	**60**	Watch
p2 early-career	2 years in, fast learner, thin network	69	73	75	38	**63**	Watch
p5 high-performer	Recently promoted PM, active brand	94	80	100	81	**88**	Strong

What this demonstrates: the scoring rubric differentiates — overall scores span 42 to 88; dimension profiles move independently (p2 and p1 share a band for opposite reasons: thin network versus stalled momentum); each persona retrieves a different career-path set; the sparse-data persona (p6, career break, tight constraints) processes cleanly; all six reports pass the citation lint; and the review workflow functions (exercised once: 6.5 minutes, on p3).

What this cannot demonstrate — stated without hedging: customer demand, usefulness to real people, willingness to pay, recommendation accuracy in real careers, real report quality, or product-market fit. Six profiles written by the developer to exercise the rubric are unit-test fixtures, not a user study. This report never counts them as customer evidence.

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7. Alternatives and the Market Gap

Alternative	Strength	Characteristic failure for this ICP	CareerArth AI's edge
Generic LLM chat (ChatGPT/Claude)	Free, fluent, always on	No structure, no grounding, quality depends on the user's prompting skill	Fixed framework, cited evidence, human review — discipline the user doesn't have to supply
Career coach	Deep human judgment	Large vague commitment before the problem is even named	Cheap structured first step; coach's time spent on strategy, not intake
LinkedIn / platforms	Market data, network	A feed, not a decision	A synthesized point-of-view with named trade-offs
Personality assessments	Established methodology	Describes traits, not this situation	Situational: uses actual role, tenure, goals, constraints
Free content	Breadth	User does all the synthesis	7 minutes to a personalised, consultant-reviewed brief

Critical honesty about the moat (developed further in §10): none of the *technology* here is defensible — the pipeline, rubric, and prompts could be replicated by a competent team in weeks. The differentiation that might endure is (a) the pairing of AI speed with genuine human review as a trust posture, and (b) CareerArth's existing brand and advisory funnel, which a standalone tool cannot copy. [HYPOTHESIS — user-perceived differentiation is untested.]

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8. Business Model

8.1 Three models compared [HYPOTHESIS]

	A. Paid AI report	B. Free diagnostic → paid consultation	C. Premium AI + human bundle
Customer trust at payment moment	Low — pay before any evidence of quality	High — value shown before money asked	Medium-high
Scalability	High (software margins)	Diagnostic scales; revenue capped by consultant hours	Capped by consultant throughput
Consultant dependency	Low	High but well-targeted (only high-intent clients)	Highest
Fit with MVP as built	Weak — the report ships in draft status before review completes	**Strong — mirrors the flow as built and the live site's motion**	Medium — needs delivery re-sequencing

8.2 Recommendation: Model B

Free (or nominal-cost) AI diagnostic as top-of-funnel; the paid human consultation (hypothesised ₹1,999–₹4,999) is the revenue event.

Four reasons, in order of weight: (1) it formalises the funnel the production CareerArth site already runs — free assessment → "Talk to a Career Expert" — so it requires no new customer behaviour and no product changes; (2) it matches the MVP's actual delivery order; (3) it avoids demanding payment for a score whose own methodology note disclaims validation — protecting the honesty-based positioning; (4) Model C remains a credible phase-2 evolution once conversion evidence exists.

8.3 Pricing and unit economics [HYPOTHESIS — every input unvalidated]

Estimated AI cost per report: **$0.05–$0.15 (~₹4–₹13)** — computed from the model's published list prices and estimated token volumes, because no live API call has ever been made. This is an estimate, not a measurement, and is labelled as such wherever cited.

Contribution model per *paying* customer (consultation price − amortised free-tier AI cost − consultant session cost − ~2.5% gateway fee):

Scenario	Price	Assumed free→paid conversion	Consultant session	
Conservative	₹1,999	5%	60 min @ ₹1,500/hr	**≈ ₹194**
Base	₹2,999	10%	45 min @ ₹1,000/hr	**≈ ₹2,089**
Upside	₹4,999	20%	30 min @ ₹800/hr	**≈ ₹4,453**

Conversion rate, prices, consultant rates, and session lengths are all assumptions.

The genuinely useful finding is structural: AI cost is 1–13% of price in every scenario — a rounding error. **Consultant time is the dominant variable cost, and conversion rate is the dominant margin lever.** The venture's economics therefore hinge on two questions a pilot can answer cheaply — does the diagnostic convert, and does the AI brief actually shorten consultant work? — and not on any AI cost optimisation.

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9. Customer Journey and Funnel

The full journey — discovery → landing → diagnostic → report → consultation interest → paid service → follow-up/referral — is mapped stage-by-stage in the business appendix. The essential points:

- **The MVP already measures the funnel's critical early stages** [DEMONSTRATED]: intake completion (with per-step drop-off location), report generation and viewing, usefulness rating (1–5 widget), and consultation-CTA clicks. A pilot reads these numbers directly from the metrics endpoint.
- **The gap is at the money end** [DEMONSTRATED as a gap]: the consultation CTA records interest but books nothing — no calendar, no payment. This is deliberate scope control, but it means "interest ≠ payment" remains unbridged evidence-wise, and the pilot's willingness-to-pay questions are the interim instrument.
- Positioning throughout the journey follows one rule: **structure, evidence, and human review are the promise — never predictive authority.** The intake page itself carries the disclaimer that scores are heuristic and unvalidated.

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10. Competition and Differentiation — the honest version

What CareerArth AI actually differentiates on today: the *combination* of a fixed diagnostic framework, deterministic scoring, per-claim evidence citation, curated labor-market grounding, and mandatory human review — wrapped in an existing advisory brand. No single element is defensible; the combination plus the brand context is the bet. [HYPOTHESIS]

What it does **not** differentiate on, stated to avoid any overclaim:

- **No technological moat.** Keyword retrieval over 18 hand-curated entries and a well-written prompt are replicable. Claimed otherwise nowhere in this report.
- **No data moat yet.** No proprietary outcome data exists; the rubric's weights are authored, not learned.
- **No validated quality edge.** Until a real Claude-generated report is produced and read (§13, day 1–30), even the quality comparison against raw ChatGPT is untested.

The durable asset, if the venture proceeds, would be built deliberately: real funnel data, consultant-review learnings, and eventually outcome tracking — none of which exist today.

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11. Go-To-Market (first 10–50 users)

Lean by design; this is a validation-stage GTM, not a marketing plan. [HYPOTHESIS]

1. **First ~10:** direct outreach to known contacts matching the ICP — doubling as pilot recruitment (recruitment message and consent text are written and ready in the validation appendix).
2. **Next ~20:** a referral ask built into every pilot conversation ("who else do you know in a similar spot?").
3. **Remaining ~20:** one or two honest "help me test this prototype" posts in career communities — explicitly not sales pitches.

University/alumni networks, partnerships, paid channels, and content are deliberately deferred: they are either mis-targeted for this ICP or premature before conversion evidence exists.

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12. Privacy, Trust, and Ethics

- **Consent is enforced in code** [DEMONSTRATED]: the pipeline refuses any profile without explicit consent; synthetic data is flagged as synthetic; real participant data (when the pilot runs) requires recorded informed consent and stays out of the repository.
- **No overclaiming by construction** [DEMONSTRATED]: hedged language and no-guarantee rules live in the generation prompt *and* the review checklist; the heuristic-score disclaimer ships inside every score object, on the intake page, and in the report footer.
- **Data minimisation:** name and email are stripped from the content sent to the model; data stays in local files; no fine-tuning, no third-party sharing.
- **Known trust limitations, openly listed** [DEMONSTRATED as limitations]: self-reported answers are unverified (the only cross-check is the human review layer); review URLs are unauthenticated (acceptable for a demo, unacceptable beyond it); an AI-drafted report reviewed by a consultant is still not licensed career counselling, and the consent text for the pilot says so.

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13. Validation: What Would Be Tested Next

13.1 The assumption register (top 5 of 10, ranked)

#	Assumption	Cheapest test	Success threshold
1	A real Claude-generated report is clearly better than the mock template	Generate 3–5 real reports; read them before recruiting anyone	Team judges them specific enough to show a stranger
2	The ICP recognises itself in the report and finds it credible	Pilot post-survey (accuracy, specificity, trust)	Averages ≥ 3.5/5
3	Free→paid conversion can sustain Model B	CTA click-rate, already instrumented	≥ 30% of report viewers express interest
4	Real users complete the 20-item intake	Completion rate, already instrumented	≥ 80%, median 5–8 minutes
5	Stated willingness-to-pay clears a viable floor	Unanchored WTP question post-report	Median ≥ ₹1,500

13.2 The pilot (designed, not executed)

10–20 participants matching the ICP: recruit → consent → pre-survey (4 questions) → real diagnostic → real report → post-survey (8 questions) → willingness-to-pay questions (4) → optional 10-minute interview → consultation-interest measurement. All quantitative steps map to instrumentation already in the MVP; the qualitative materials are fully drafted. **No new engineering is required to run it.** If any result is within ~10% of its threshold at this sample size, it is treated as inconclusive and the interview data governs.

13.3 The 30/60/90-day roadmap

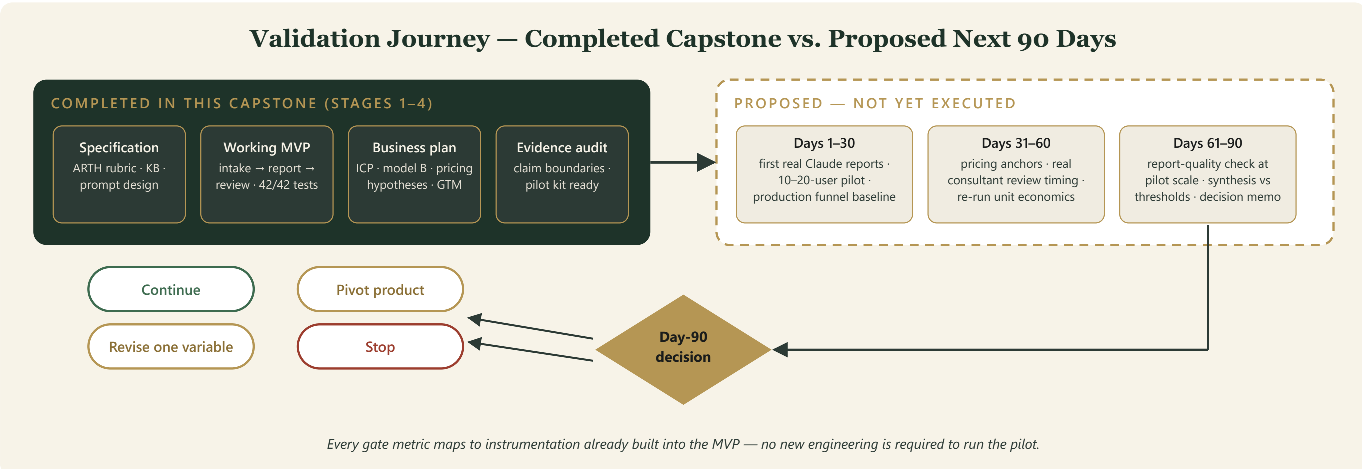

Figure 3 — Validation journey: completed capstone stages versus the proposed 30/60/90-day experiment and decision gate.

- **Days 1–30:** first real Claude generations (read before recruiting); run the pilot; pull the production funnel's real baseline conversion — the cheapest high-leverage data point available.
- **Days 31–60:** pricing-anchor testing; time a real consultant reviewing a real report; re-run unit economics with measured (not estimated) AI cost and review time; re-decide Model B

vs. A/C on evidence.

- **Days 61–90:** report-quality and groundedness checks at pilot scale; synthesis against thresholds; a written **continue / revise / pivot / stop** decision memo citing specific numbers. Explicit stop condition: low completion *and* low interest *and* generic-reading real reports means the core value proposition — not the packaging — has failed.

No product features are added anywhere in this roadmap until the day-90 decision is "continue."

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14. Limitations

Consolidated and unhedged:

1. **No real user has ever used the product.** All personas are synthetic; all survey instruments are unexecuted.
2. **No real AI report has ever been generated.** All reports are mock-template output; the flagship capability is implemented but unobserved.
3. **The ARTH rubric is heuristic and partly circular** — calibrated to reproduce the site's marketing sample, not validated against outcomes. Every score carries this disclaimer.
4. **All pricing, conversion, and cost-side figures are assumptions or estimates**, and are labelled as such at every appearance.
5. **The knowledge base covers 18 roles in 3 industries**; users outside it get a degraded (though honestly flagged) experience.
6. **Self-report bias is unmitigated in-product**; the human review layer is the only check.
7. **The MVP is a prototype, not a production system**: single-process, file-backed, unauthenticated review URLs, no booking or payment.
8. **The production funnel's real conversion baseline was never measured** — its *shape* supports Model B; its performance is unknown.

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15. Conclusion

This capstone set out to answer an entrepreneurial question with an unusually honest evidence posture: build the real product, model the real business, and then refuse to blur the line between what has been demonstrated and what remains a hypothesis.

What stands demonstrated is meaningful: a complete, tested, instrumented product flow that turns a seven-minute structured intake into a scored, evidence-cited, humanly reviewable career diagnostic — architected so that its weakest scientific point (a heuristic rubric) is contained, disclosed, and auditable rather than hidden behind AI fluency. What stands unvalidated is equally clear, and this report has named it repeatedly rather than once: no real user, no real report, no real rupee of evidence on conversion or willingness to pay.

The venture's next step is therefore not a build decision but an evidence decision, and it has been made executable: a pilot that needs no code, materials ready to send, and thresholds committed to in writing *before* the data arrives — so that in ninety days, "continue, revise, pivot, or stop" can be answered by numbers against promises, which is the discipline this course teaches and the discipline this document has tried to practice.

Appendices: Business & Validation Appendix (04), Technical & Product Appendix (05), Pitch Deck (03), Demo Guide (06). All source material: `careerarth-ai/docs/` *Stages 1–4.*